

YEAR 9 NUMERACY – SET 5

Use a calculator to help answer these questions.

Name Date

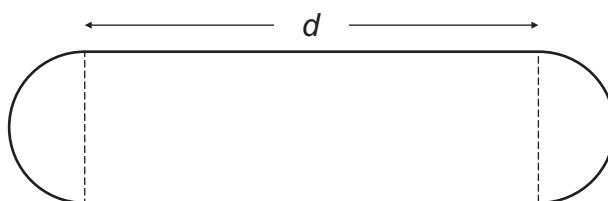
A quadrilateral has two equal angles that are obtuse.

One of the other angles is 50° .

Which of these statements is **definitely true** about the fourth angle?

- 1
- ☐ It is obtuse.
 - ☐ It is less than 130° .
 - ☐ It is more than 130° .
 - ☐ It is either acute or right.

The diagram shows a 400-metre running track.



- 2 To find the radius of the semicircular ends of a track like this,

you use the formula $r = \frac{200 - d}{\pi}$

where r is the radius and d is the length of the straights as marked in the diagram.

If the straights are each 75 m long, the radius is closest to

- ☐ 16 m. ☐ 40 m. ☐ 152 m. ☐ 176 m.

Which of these four countries has the highest population density (in people per km^2)?

3

Country	Area (km^2)	Population
Costa Rica	51 100	4 016 200
Egypt	1 001 450	77 505 800
Greece	131 940	10 668 400
Spain	504 782	40 341 500

- ☐ Costa Rica ☐ Egypt ☐ Greece ☐ Spain

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A bookshop has a sale.

A book normally costing \$39.95 is priced at \$23.97.

- 4 By what percentage has the price been reduced?

☐ 16%

☐ 40%

☐ 60%

☐ 67%

5

When $k = 3$ and $p = -1$ what is the value of $\frac{2k^2 + 3}{5 - 2p}$?

☐ 3

☐ 5

☐ 7

☐ 13

The win : loss ratio of an AFL team after playing 15 games was 2 : 3.

The team then won four of the last seven games of the season.

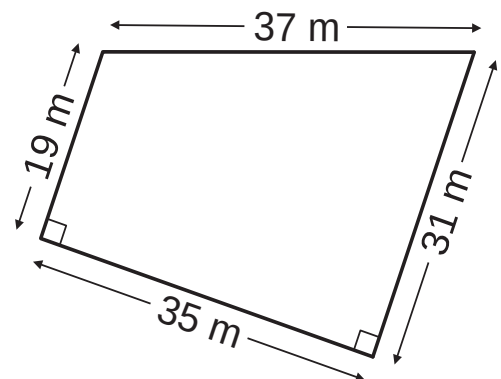
- 6 What was the team's win : loss ratio at the end of the season?

:

What is the area of this block of land?

7

m²



8

Which one of the following composite numbers has **no** prime factors less than 20?

☐ 25781

☐ 25783

☐ 25787

☐ 25789